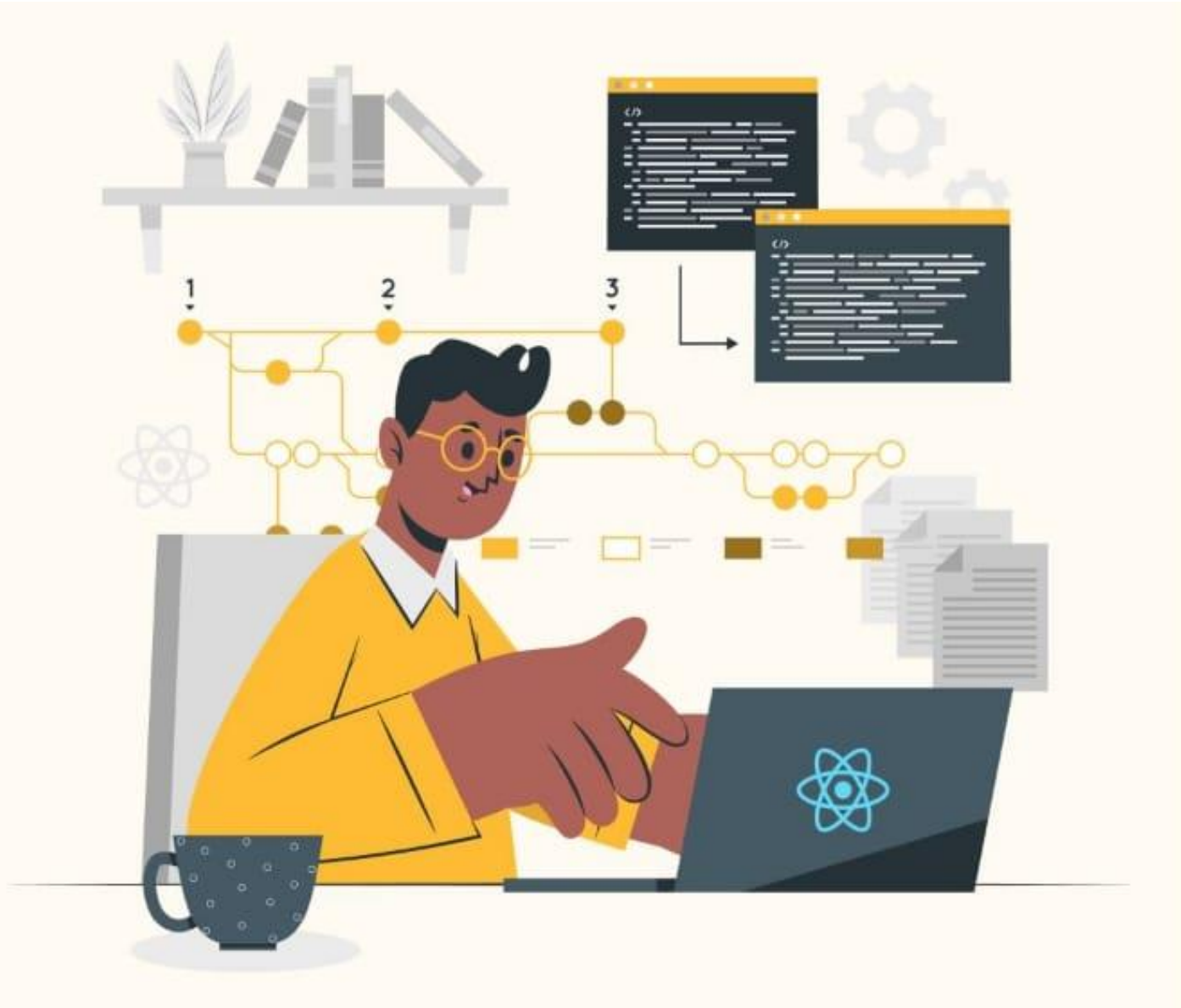




An Introduction To DATA SCIENCE & MACHINE LEARNING

By: Dipak Patel

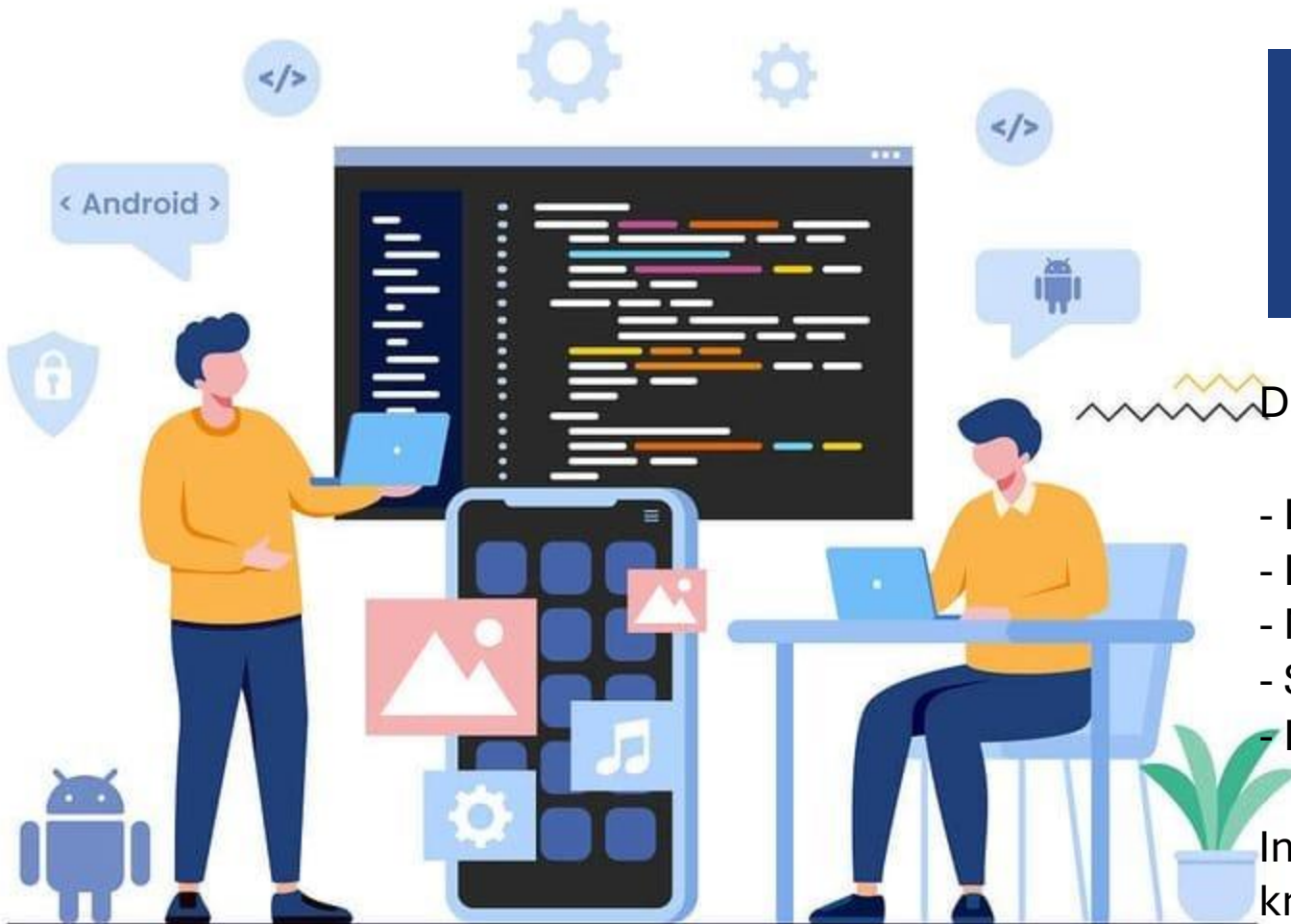
What Is Data science ?

Data science: Extracting insights and knowledge from data to inform decisions.

Data science is used to:

- Extract insights from data
- Identify patterns and trends
- Make predictions and informed decisions
- Solve complex problems
- Drive business value and growth

In short, data science helps turn data into actionable knowledge!



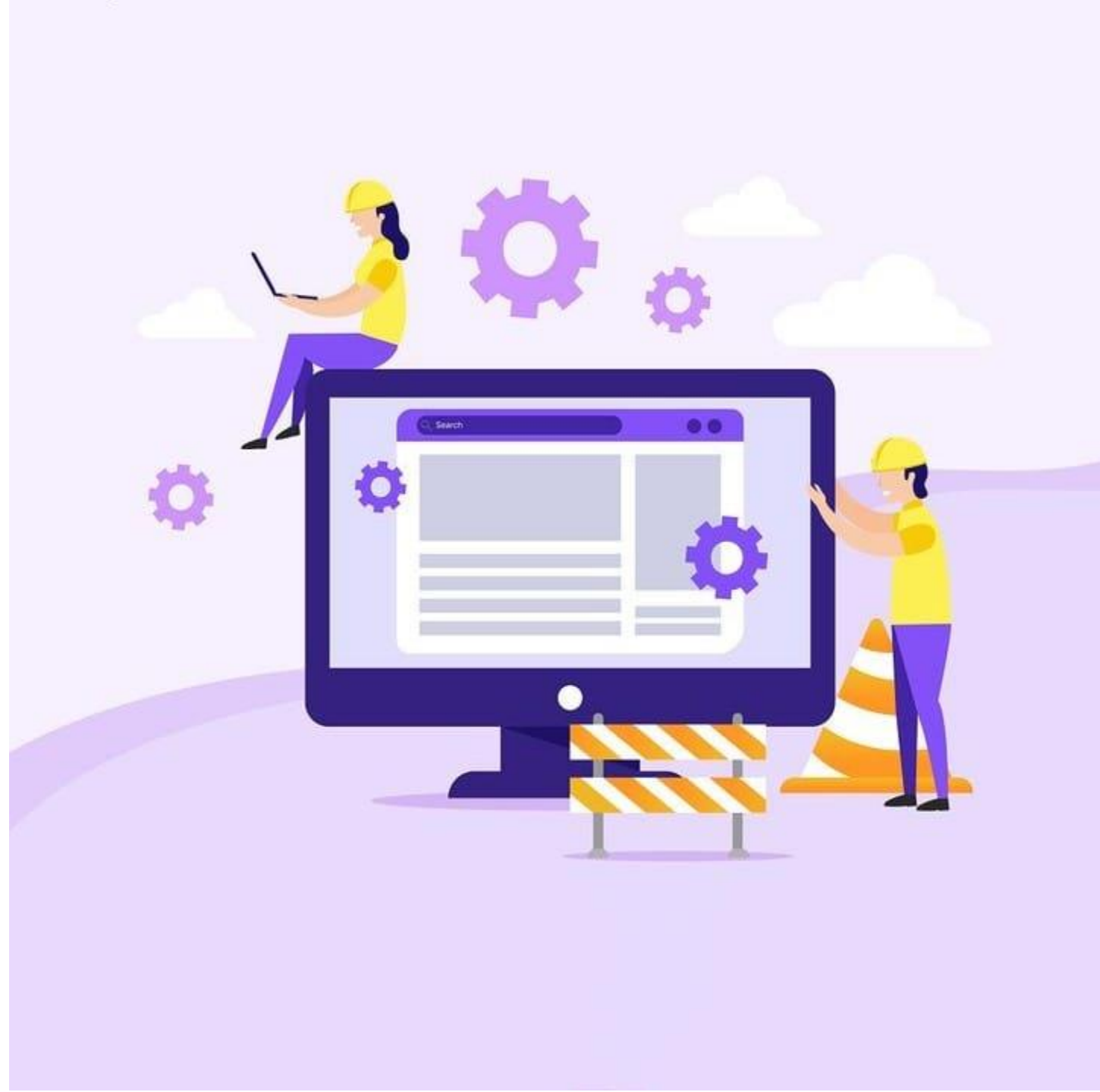
➤ TABLE OF CONTENTS

DATA SCINCES

ML

Data Analytics

Big Data Analytics



Basic Of Data Science





➤ There are 4 step-by-step procedures for learning the data science process

- 1) Learn A Programming Language
- 2) Learn Some Statistics
- 3) Data Visualizations
- 4) Projects

1) Learn A Programming Language



- 1) Python
- 2) R

{Python}

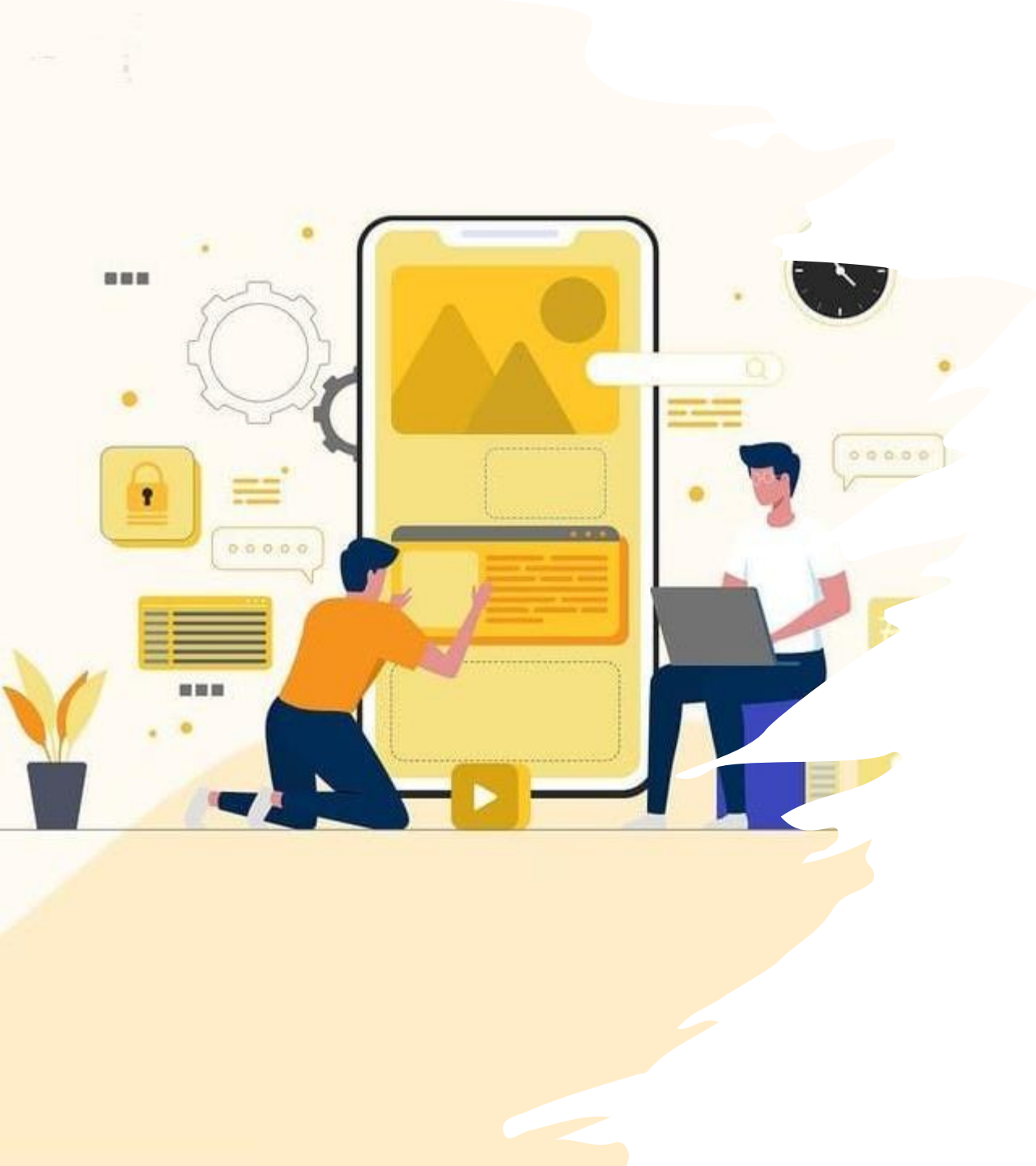
- 1) Variable
- 2) If else
- 3) Loops

- 1) NumPy Library
- 2) Pandas Library

2) Learn some statistics

- 1) statistics
- 2) Linear algebra
- 3) calculus
- 4) probability





3) Data Visualization

- Matplotlib library
- Seaborn library

4) projects

A project is defined as a sequence of tasks that must be completed to attain a certain outcome.



MACHINE LEARNING



WHAT IS MACHINE LEARNING

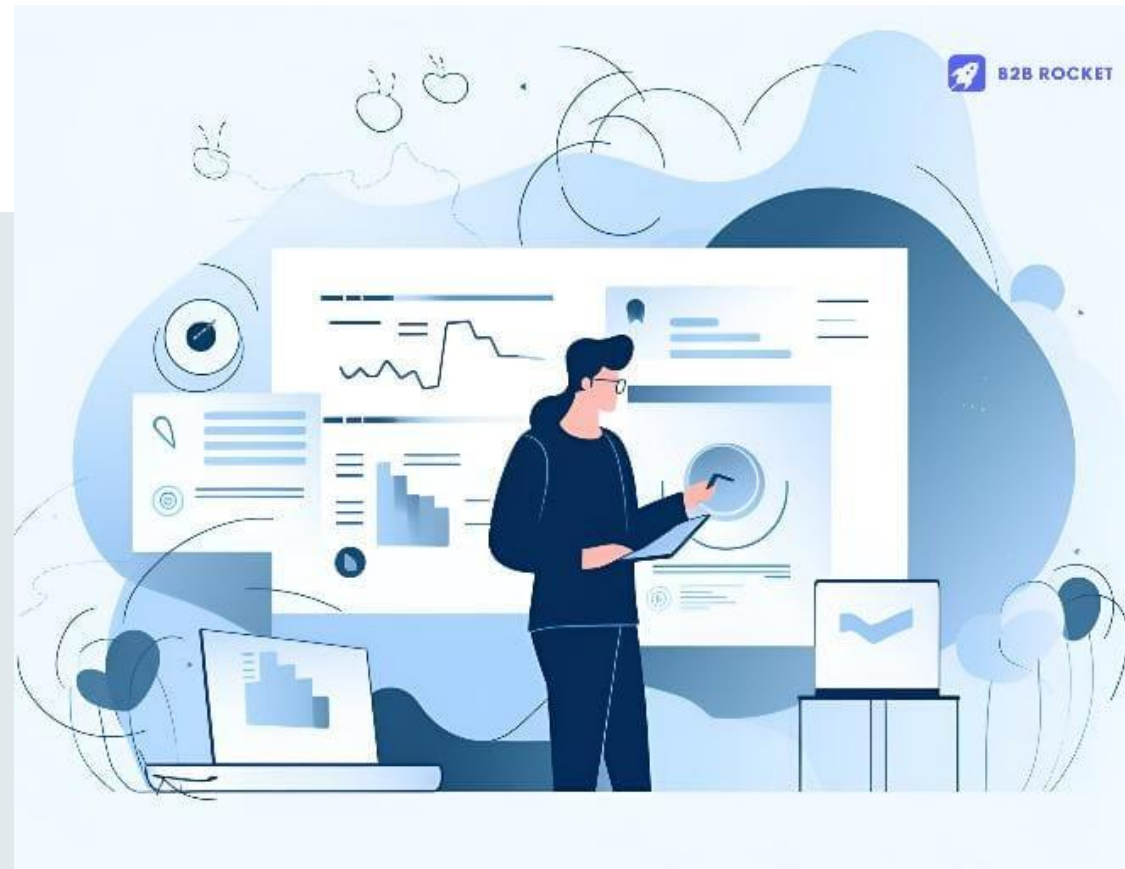
- machine learning enables computers to learn and improve automatically, without being explicitly programmed!



Basic of machine learning

➤ There are 3 step-by-step procedures for learning the machine learning process

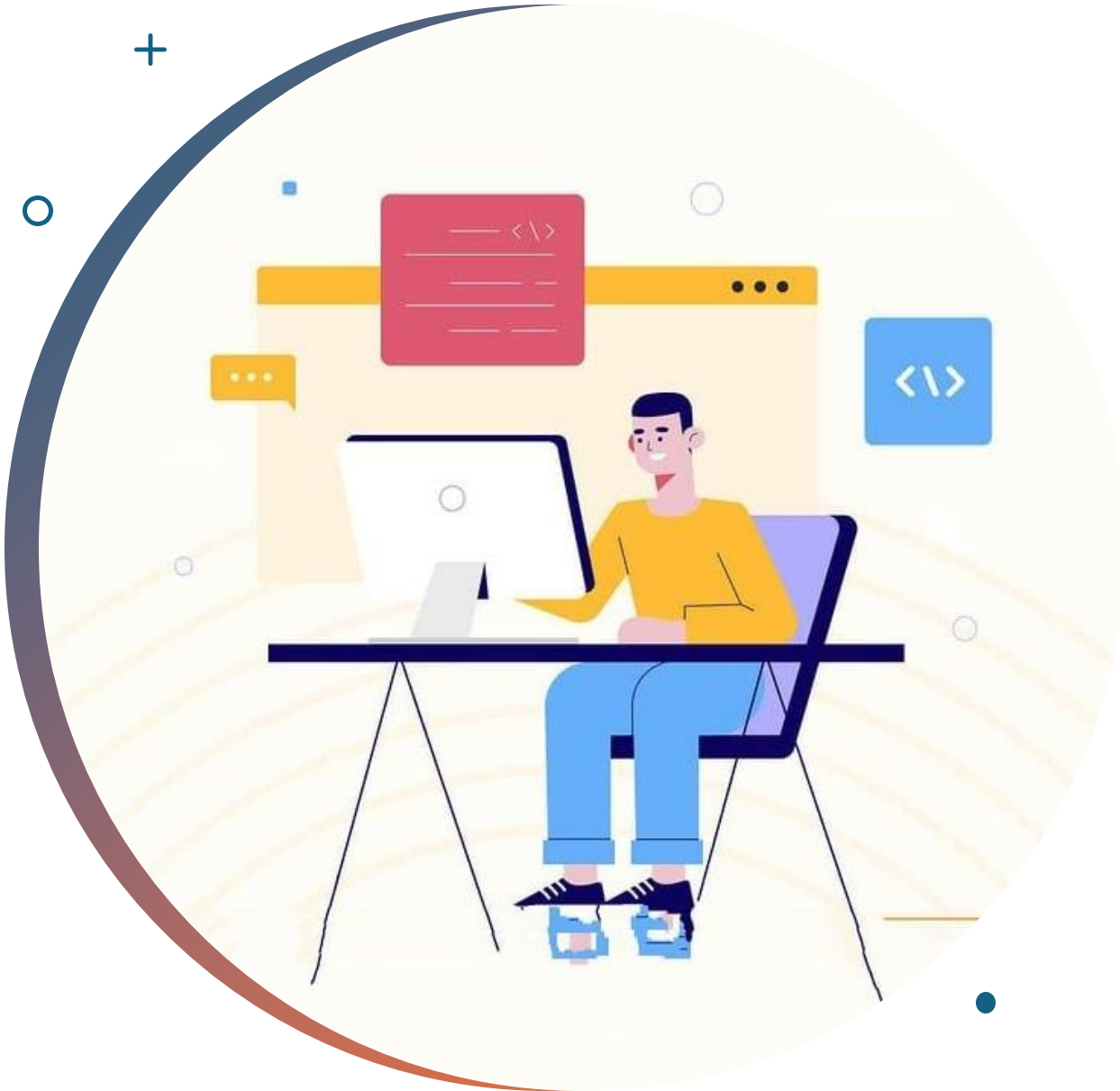
- 1) Learn ML Algorithms
- 2) Data Preprocessing
- 3) Learn ML Libraries



1) Learn ML Algorithms

- Supervised vs Unsupervised vs Reinforcement
- Linear Regression, Logistic Regression, Clustering
- KNN (K Nearest Neighbours)
- SVM (Support Vector Machine)
- Decision Trees
- Random Forests
- Overfitting, Underfitting
- Regularization, Gradient Descent, Slope
- Confusion Matrix





2) Data Preprocessing (for higher accuracy)

- Handling Null Values
- Standardization
- Handling Categorical Values
- One-Hot Encoding
- Feature Scaling

3) Learn ML libraries

- Scikit learn
- Matplotlib
- Tensorflow for DL





Thank You